Supplementary Material for "Does Splitting Returns with Wavelets Help GARCH Forecast Volatility?"

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1. Code

All code, data, configuration, tests and result tables are here:

https://github.com/pushpendras0026/Volatility-forecasting-with-the-wavelet-transform

The repository README explains how to run everything with one command (`Rscript run_all.R --workers 16`). It needs only R 4.6 with the recommended package `nnet`, no internet. The run first does the unit and leakage tests, then all forecasts, then the tables and figures, then the innovation-distribution experiment. The recorded model-fitting time of the final run is about 3.8 CPU-hours in total, spread over the workers. The bulky raw forecast files are stored as zip files in the repository and only need to be unzipped if you want to re-make the tables without re-running the models.

2. How to implement the method

This section describes the full procedure in words and pseudocode, so somebody can re-build it without our code.

2.1. Data preparation

1. Read daily closing prices, sort by date, drop duplicate dates. Do not fill missing days.
2. Compute $r_t = \ln P_t - \ln P_{t-1}$ (decimal, not percent).
3. Compute the target $y_{t+1} = (r_{t+1} - \bar{r}_t^{(20)})^2$ where $\bar{r}_t^{(20)}$ is the mean of $r_{t-19}, \ldots, r_t$.
4. Split by calendar: training up to 31 Dec 2009, validation 2010–2011, test 2012–2014, later period 2015–2025 (stocks only).

2.2. Rolling forecast loop

Algorithm 1 Rolling one-step forecast for one model

Input: returns $r_{1..T}$, first forecast origin t_0, window $W = 1024$, refit cadence $K = 20$, frozen order (p, q)
for $t = t_0$ **to** $T - 1$ **do**
  if $(t - t_0)$ mod $K = 0$ or last fit failed **then**
    take the last W observations up to t
    transform them (raw / paired DWT / MODWT)
    fit ARMA(p, q) by Gaussian ML
    fit GARCH(1,1) on its residuals by ML
  else
    keep parameters, update ARMA state and h_t with the new observation
  end if
  get one-step μ and h of the input series
  map them to return moments μ_{t+1}, h_{t+1}
  $\hat{y}_{t+1} = h_{t+1} + (\mu_{t+1} - \bar{r}_t^{(20)})^2$
end for

A failed fit is tried again at the next day; we never fall back silently to an old fit, and the failed day is marked missing. Comparisons then use only dates where all compared models have a forecast.

2.3. Order selection

On the last 1,024 training observations, fit every ARMA(p, q) with $0 \le p, q \le 3$ and keep the one with the lowest AIC $= -2\ell + 2k$, where k counts the ARMA coefficients, the constant and the variance. This order is then frozen for validation and test. For the paired DWT, the order is chosen separately for the approximation and the detail series. Table 1 lists the chosen orders and the GARCH parameters of the first fit.

2.4. GARCH likelihoods

Given ARMA residuals ε_t, the variance follows $h_t = \omega + \alpha\varepsilon_{t-1}^2 + \beta h_{t-1}$, started at the sample variance. We maximise the log-likelihood with box constraints $\omega > 0$, $\alpha, \beta \ge 0$, $\alpha + \beta < 1$:

- Gaussian: $\ell = -\frac{1}{2}\sum[\ln(2\pi h_t) + \varepsilon_t^2/h_t]$.
- Student-t with $\nu > 2$, scaled to unit variance: $z_t =$

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Table 1. ARMA orders chosen by AIC and the first GARCH(1,1) fit at the end of training (Gaussian). DWT has two series: detail (d) and approximation (a).

Series	Model	(p, q)	α	β
S. Africa	raw	(3,1)	0.113	0.873
	DWT d / a	(2,2) / (3,3)	0.189 / 0.116	0.784 / 0.855
	MODWT	(3,2)	0.116	0.868
Tunisia	raw	(2,0)	0.166	0.760
	DWT d / a	(2,3) / (1,2)	0.244 / 0.184	0.631 / 0.619
	MODWT	(2,1)	0.164	0.761
Nigeria	raw	(0,3)	0.155	0.790
	DWT d / a	(3,0) / (0,1)	0.207 / 0.118	0.566 / 0.881
	MODWT	(3,1)	0.152	0.791
Infosys	raw	(0,3)	0.108	0.872
	DWT d / a	(3,2) / (2,2)	0.169 / 0.063	0.773 / 0.926
	MODWT	(2,2)	0.115	0.865
IBM	raw	(2,2)	0.079	0.904
	DWT d / a	(3,3) / (2,2)	0.096 / 0.082	0.881 / 0.894
	MODWT	(2,3)	0.077	0.906
Google	raw	(3,2)	0.049	0.948
	DWT d / a	(2,2) / (1,1)	0.073 / 0.062	0.922 / 0.920
	MODWT	(0,1)	0.050	0.947

$\varepsilon_t/\sqrt{h_t}$ has density $\frac{\Gamma(\frac{\nu+1}{2})}{\Gamma(\frac{\nu}{2})\sqrt{\pi(\nu-2)}}\big(1+\frac{z^2}{\nu-2}\big)^{-\frac{\nu+1}{2}}$, minus $\frac{1}{2}\ln h_t$.

- GED with shape κ, unit variance: density $\frac{\kappa\exp(-\frac{1}{2}|z/\lambda|^\kappa)}{\lambda\,2^{1+1/\kappa}\Gamma(1/\kappa)}$ with $\lambda=\big(2^{-2/\kappa}\Gamma(1/\kappa)/\Gamma(3/\kappa)\big)^{1/2}$. $\kappa=2$ is Gaussian and $\kappa<2$ has fatter tails.

2.5. Wavelet transforms

Haar MODWT, level 1. $w_t=(r_t-r_{t-1})/2$, one value per day. Given (μ_w, h_w) for w_{t+1}, $\mu_r=r_t+2\mu_w$, $h_r=4h_w$.

Paired Haar DWT. Make pairs from the newest complete pair backwards, $(u_k, v_k)=(r_{t-2j-1}, r_{t-2j})$ ending with (r_{t-2}, r_{t-1}) (an odd leftover at the oldest end is dropped). Compute $a_k=(u_k+v_k)/\sqrt{2}$ and $d_k=(v_k-u_k)/\sqrt{2}$, fit one ARMA–GARCH to each, and combine with the conditional formulas in Section 3.3 of the main report, where $s=\sqrt{2}\,r_t$ is the known part of the next pair.

Component models (E3). At each origin, decompose the observed history of y (DWT or MODWT, Haar/db4/db6, $J=1,2$) with reflection at the edges, so only past data is used. Forecast each of $A_J, D_J, \ldots, D_1$ one step ahead and add the forecasts; clip the final sum at zero. Figure 1 shows a Haar MODWT split with $J=2$.

2.6. Ensembles and MLP

Ensemble forecasts are the plain average of the member forecasts on the same day. If any member is missing that day, the ensemble is missing too (we do not average fewer members). The MLP uses inputs $(y_t, \ldots, y_{t-4}, \bar{y}_t^{(5)}, \bar{y}_t^{(20)})$,

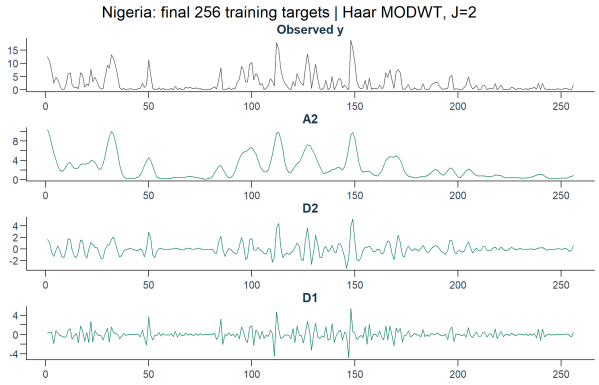


Figure 1. Nigeria, last 256 training targets ($\times 10^4$) and their Haar MODWT parts with $J=2$. The parts add up to the observed series.

Table 2. Sensitivity: MSE gain (%) of each setting change vs. the default (window 1024, refit 20, AIC, cap 3), raw GARCH, test 2012–2014. Pairwise matched dates.

Change	SA	Tun	Nig	Inf	IBM	Goog
Window 768	1.15	−14.36	1.77	−0.19	0.36	0.08
Window 1536	−0.26	7.93	7.19	0.36	−1.98	−1.61
Refit 40	0.40	0.02	−1.55	−0.07	−0.27	0.06
Refit 60	0.37	2.31	−1.76	−0.64	−0.75	0.01
BIC	−0.10	0.34	0.00	1.23	0.07	0.26
Order cap 5	−0.32	−2.56	0.07	−0.32	−1.02	0.10

standardised with training means and standard deviations, one hidden sigmoid layer, a linear output, at most 300 iterations; fits that hit the iteration limit are marked as unavailable.

3. Extra results

Tables 2–4 give more detail on the results in the main report. In Table 4 the number of dates is small for Infosys (199) because some MLP and ensemble forecasts were missing on many days; raw GARCH RMSE on these dates is different from the main table only because the dates are different.

Matched-date rule. Every number in our result files comes with a `sample_rule` column that says which dates were used. The file `tables/sample_reconciliation.csv` in the repository shows the same model under every rule, so the small differences between tables can be checked.

Table 3. Test RMSE $\times 10^4$ (2012–2014) for DWT and MODWT pipelines under the three innovation distributions. Dates where all three distributions have a forecast.

Series	Pipeline	Gaussian	Student-t	GED
S. Africa	DWT	1.3448	1.3448	1.3448
	MODWT	1.3394	1.3389	1.3391
Tunisia	DWT	0.6790	0.6564	0.6622
	MODWT	0.7061	0.7013	0.7023
Nigeria	DWT	1.8015	1.8090	1.8042
	MODWT	1.9173	1.9073	1.9129
Infosys	DWT	24.9339	24.8529	24.9561
	MODWT	25.1997	25.0508	25.1189
IBM	DWT	4.1669	4.1766	4.1679
	MODWT	4.1395	4.1400	4.1367
Google	DWT	7.7759	7.7907	7.7822
	MODWT	7.7684	7.7564	7.7489

Table 4. MLP seeds vs. raw GARCH, test RMSE $\times 10^4$, on dates where raw GARCH, all three MLP seeds and the three ensembles have a forecast.

Series	n	Raw	Seed 101	Seed 202	Seed 303
S. Africa	688	1.3771	1.5194	1.5140	1.4648
Tunisia	540	0.3794	0.4083	0.4201	0.5413
Nigeria	745	1.8920	2.4044	2.0768	2.0256
Infosys	199	23.1670	33.2734	24.0858	33.2707
IBM	751	4.1300	4.2190	4.1655	4.2011
Google	704	7.9898	8.1453	8.0270	8.0803